

Neuromorphic-Enabled Implementation of Extremely Low-Power Gaze Estimation

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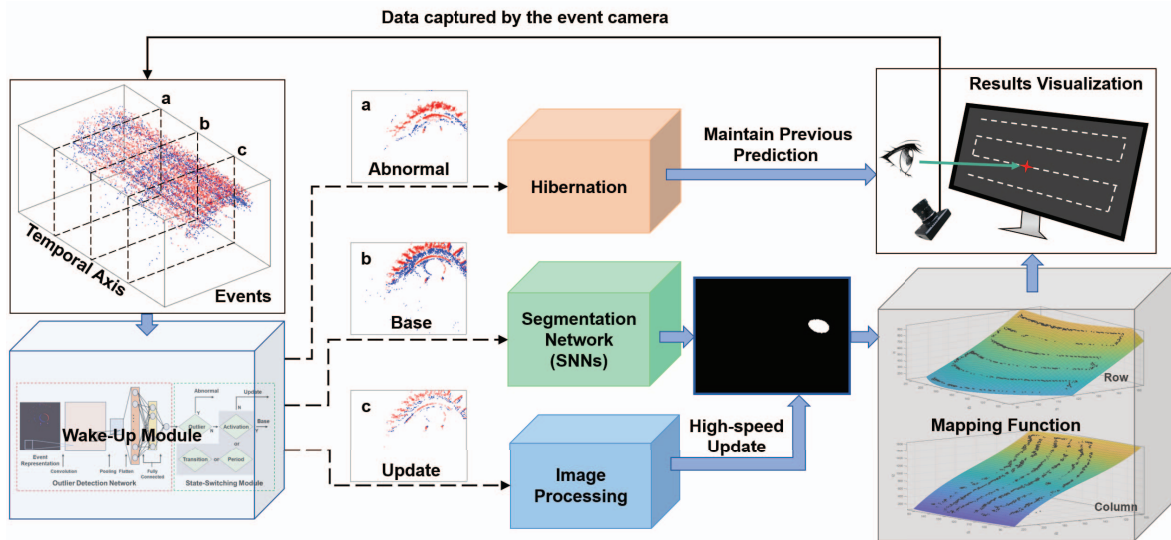


Figure 1: Workflow of event-based low-power spiking gaze estimation architecture.

ABSTRACT

Event camera has great potential in the field of eye tracking, while current event-based gaze estimation suffers from complex imaging settings and participation of RGB modality. We propose a novel architecture for completely event-based low-power spiking gaze estimation using only one eye signal. Our architecture employs a wake-up module to judge the state of inputs, and then enters one of the three modules including hibernation, lightweight SNN segmentation network, and image processing module to obtain the gaze estimation results. We prove that our method performs better in terms of accuracy and power consumption on Angelopoulos's dataset.

Keywords: Gaze estimation, event camera, wake-up module, spiking neural network

Index Terms: Human-centered computing—Human computer interaction (HCI)—HCI theory, concepts and models

The event camera is a novel kind of neuromorphic vision sensor, with notable advantages of high temporal resolution (microseconds), high dynamic range (>120dB), and no motion blur, showing great

potential in the field of eye tracking. At present, there have been some successful attempts to apply event cameras to eye tracking tasks. A hybrid frame-event-based eye tracking system is realized in [1], yet it still requires images generated by conventional cameras. Later, a purely event-based method for detecting corneal glints is proposed in [3]. However, this method requires complex illuminating equipment to assist the system. Then, a real-time eye tracking algorithm is presented in [2], while the working frequency is only about 30 Hz.

In this paper, we propose a fully event-based method for eye tracking as shown in Fig. 1, which can achieve low power consumption and high-speed refresh without using complicated equipment. We propose a wake-up module that can switch appropriate modes based on the inputs, achieving a balance between accuracy and computational complexity. We propose a lightweight spiking neural network for semantic segmentation, further reducing the power consumption.

1 METHOD

The proposed architecture consists of a wake-up (WU) module, a segmentation network, an image processing module, and a mapping function, as shown in Fig. 1. The event stream will first be converted into event frames before entering the WU module. The WU module is used to switch the corresponding mode according to the input event frames, and a segmentation network or an image processing module will extract the pupil region from the event frames. The mapping function will further calculate the gaze point based on the pupil region.

1.1 Wake-Up Module

Easy wake-up and low power consumption are the merits of our system, which are dependent on the wake-up (WU) module. The

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WU module consists of an outlier detection (OD) network and a state-switching (SS) module. The OD network consists of a convolutional layer, a pooling layer, and two fully connected layers, with the final sigmoid activation function mapping values to a range between 0 and 1. The SS module first rounds the network output value to 0 or 1 to determine whether the input data is an outlier. If the input data is an outlier, the system will enter *abnormal* mode; otherwise, it will enter *base* mode or *update* mode, which is determined by three parallel conditions, i.e., transition condition, activation condition, and period condition.

1.2 Segmentation Network

Inspired by the structure of Squeeze U-Net, we propose a lightweight network. Our network consists of down-sampling units in the contracting path, and up-sampling units in the expansive path. The extracted features are sent to the next down-sampling unit and the corresponding up-sampling unit. In order to further reduce the computational load of the network, we swap the neurons in ANNs for spiking neurons to create SNNs. We use spatio-temporal backpropagation (STBP) [4] to train our SNNs here. Eye segmentation is a common method to extract eye features for model-based gaze estimation. Considering the factors of model robustness and update frequency, we finally choose pupil segmentation for gaze estimation. The pupil region extracted from the image is used as the label for network training.

1.3 Image Processing

In most cases, the segmentation can be updated rapidly rather than re-calculated by the segmentation network, since the location of the pupil does not change dramatically. The image processing module serves to achieve high-speed updates at an ultra-low computational cost by combining the previous segmentation knowledge and the latest events. In this study, under infrared (IR) illumination, the pupil appeared as a black ellipse with a distinct edge compared to other eye structures. Hence, the events can reveal the direction of pupil movement. Through simple morphological operations, rapid updates to segmentation results can be achieved at an extremely low computational cost.

1.4 Mapping Function

This module serves to map the pupil region to the gaze point coordinates or gaze direction vector. Considering the computational complexity, we simplify the calculation by choosing the pupil center as the only descriptor. We choose the most intuitive and concise polynomial function, and our experiments prove that the third-order polynomial achieves a good balance between fitting accuracy and formula robustness. The R-square values of the row and column coordinates are 0.9947 and 0.9968, respectively, which demonstrates the fitting is excellent.

2 EXPERIMENT

To compare with Angelopoulos's work [1], we evaluate our method using their published datasets. However, our method only uses event modality from a single eye, rather than event modality and image modality from two eyes.

As with [1], we calibrate half of the points and report the results for all the points. The quantitative comparison of accuracy between Angelopoulos's method and ours is shown in Table 1. Under the same experimental conditions, the accuracy of our method has significantly improved compared to Angelopoulos's method in both smooth pursuit and random saccades modes.

The power consumption of the system includes hardware power consumption and software power consumption. The hardware power consumption mainly comes from the system's sensors, and our estimation results indicate that the chip power consumption of Angelopoulos's system is 320 mW, while ours is 20 mW. Considering

Table 1: Quantitative Comparison of Accuracy between Angelopoulos's Method and Ours

Method	FOV (deg)	Smooth pursuit			
		Average error		Best error	
		Angle (deg)	Coordinate (px)	Angle (deg)	Coordinate (px)
[1]	113	3.9	-	-	-
Ours	113	1.9	38	1.27	25
Method	FOV (deg)	Random saccades			
		Average error		Best error	
		Angle (deg)	Coordinate (px)	Angle (deg)	Coordinate (px)
[1]	113	-	-	1.75	-
Ours	113	2.5	49	1.5	29

¹ The best error refers the smallest error among all subjects, and the average error refers the average error across all subjects.

² The symbol "-" indicates that these data are not available.

the software power consumption, the power consumption of Angelopoulos's algorithm is 0.08 mW, while ours is 82.2 mW. Since the hardware power consumption occupies the main part of the system power consumption, our proposed architecture (102.2 mW) is superior to Angelopoulos's architecture (320.08 mW) in terms of overall power consumption.

3 CONCLUSION

In this paper, we propose a fully event-based low-power spiking gaze estimation method with no need for dense image modality or complex illumination. We incorporate a WU module into the system, which allows for the majority of the data to be updated simply by image processing. Only a tiny portion (3.57%) of the data needs to be recalculated through the segmentation network, which is a lightweight SNN. According to our estimation, the overall power consumption of our system is 102.2 mW, which is only 31.9% of Angelopoulos's system (320.08 mW). Despite the modal decrease of the input data, our method achieves better accuracy, with the average angle error of 1.9° under smooth pursuit mode and 2.5° under random saccades mode. We believe that our proposed neuromorphic-enabled extremely low-power architecture will further promote the application of eye tracking in the field of AR/VR.

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